Minor Project

Project Title: Spam Email Detection Using Machine Learning

1. Introduction

Spam emails are unwanted messages that are sent to many users. These emails may contain advertisements, fake offers, or harmful links. This project builds a system that can automatically detect whether an email is **spam or not spam** using machine learning.

2. Objective

The objective of this project is to develop a machine learning model that can classify email messages such as **Spam** or **Not Spam (Ham)** based on their content.

3. Hardware Requirements

- Computer or Laptop
- Minimum 4 GB RAM
- Processor: Intel i3 or above

4. Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python

Libraries Used

- Pandas

- NumPy
- Scikit-learn
- NLTK

Tools

- Jupyter Notebook / Google Colab / VS Code

5. Functional Requirements

The system should be able to:

1. Accept an email message as input
2. Clean and process the text data
3. Convert text into numerical data
4. Train a machine learning model
5. Predict whether the email is spam or not
6. Display the result to the user

6. Non-Functional Requirements

- The system should give fast results.
- The system should be easy to use.
- The model should provide good accuracy.

7. Output

The system will classify the email as:

- **Spam**
  or
- **Not Spam**

Example:

Input:
"Congratulations! You won a free prize."

Output:
Spam